Predicting streetcar bunching risk-events from risk-patterns:

An ML approach to congestion on the TTC

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GitHub link: [elhyc/predicting-streetcar-bunching-risk-events: Predicting streetcar bunching risk-events]

Elliot Cheung

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Overview

This project develops a flexible prediction tool for predicting bunching risk-events, from confirmed bunching risk-patterns. In this project a bunching risk-event can be specified by 4 integers (**b**,**s**,**n**,**m**). Then, the bunching risk-event prediction task is given by:

- Given a number of bunching incidents ≥ **b**, observed by a pair of streetcars in the last **s** stops, predict if the same pair of streetcars will encounter another number of bunching incidents ≥ **n** within the next m stops

For any risk-event, we call a collection of bunching incidents satisfying the conditional statement of the associated prediction task, a **risk-pattern.** Then, the prediction task for a type of risk-event is a binary classification problem: for each associated risk-pattern, assign a 1 if the risk-pattern extends to a risk-event, assign a 0 if it does not.

The majority of the data used in this project was scraped from www.Transsee.ca. Transsee is a website that aggregates real-time AVL/GTFS transit data, and organizes this data into convenient charts and tables. This real-time data is publically shared by the TTC, and can be realistically acquired in real-time if this prediction tool is to be used in a realistic deployment context.

Apart from the aggregated transit data, the model also considers potentially relevant external factors such as: weather, traffic signals, pedestrian crossings and active road permits. All of this data can be found on Open Data Catalogue - City of Toronto Open Data Portal. The weather data can be found on climate.weather.gc.ca.

Use-cases and Stakeholders

In November 2025, a report (Bunching and Gapping Pilot Project Update) sent to the TTC board indicated that the issue of bunching and gapping in the TTC can be improved with a surplus of route supervisors directed towards managing the operations of each route. However, it was found that such a level of supervision is probably not scaleable to the entirety of the TTC transit system. Therefore, the development of a project like this would be of great interest to the TTC, as well as other possible transit systems that suffer through a strong congestion symptoms.

Therefore, one can potentially use predictive models such as the ones provided by this project, to aid in these efforts. Ideally, such a risk-flagging prediction tool could be part of a larger supervisory framework that takes predictive flags into account, and generates actionable solutions based on these predictions. Furthermore, when off-line, one can train predictive models arising from our project against a variety of risk-event type tasks – so

that pre-trained models are ready for deployment. These deployed prediction models should be periodically and consistently trained on more data, as time goes on.

Key Performance Indicators (KPIs)

As the goal of this project is primarily to design a prediction tool to be used for flagging potential bunching risk-events given confirmed bunching risk-patterns, a primary metric of interest here for evaluating model success is **recall rate**. We also pay attention to metrics of **precision**, and **false positive rate**. The significance of the latter two metrics (precision and false positive rate) may potentially scale with the manner in which these bunching risk-event predictions would be used in a route supervisory framework, together with a specific choice of response to the the provided predictions. Furthermore, depending on the specified risk-event type, risk-patterns classification types (0 or 1) may be skewed towards the positive class. In such a scenario, it is more meaningful to balance precision and recall, as false positives become more of a significant issue.

Feature engineering

From the raw stop-level data, significant feature engineering is required to define the targets and features for this classification task. Clearly, bunching episodes and **risk-patterns** are not immediately available from the raw stop-level dataset, but they can be extracted from the raw stop-level dataset. A **bunching incident** (at a stop) occurs between a pair of streetcars if the two streetcars in the pair arrive at the stop in a bunched manner (i.e. gap is less than some threshold defined at the outset).

A **bunching episode** between a pair of streetcars is defined to be a continuous, consecutive (both in the stop order and in time order) chain of bunching incidents experienced by the same pair of streetcars. Each bunching episode has an episode start, which is the first stop of the chain experiencing the bunching incident.

Given a bunching incident, the bunching incident in question can only be part of a risk-pattern of type (**b**,**s**,**n**,**m**) if enough bunching incidents also occur between the stop of the current incident and **s** stops prior. When a bunching incident is logged as an episode, these lookback conditions are checked, to see if the episode should obtain a "risk-pattern" or eligibility flag.

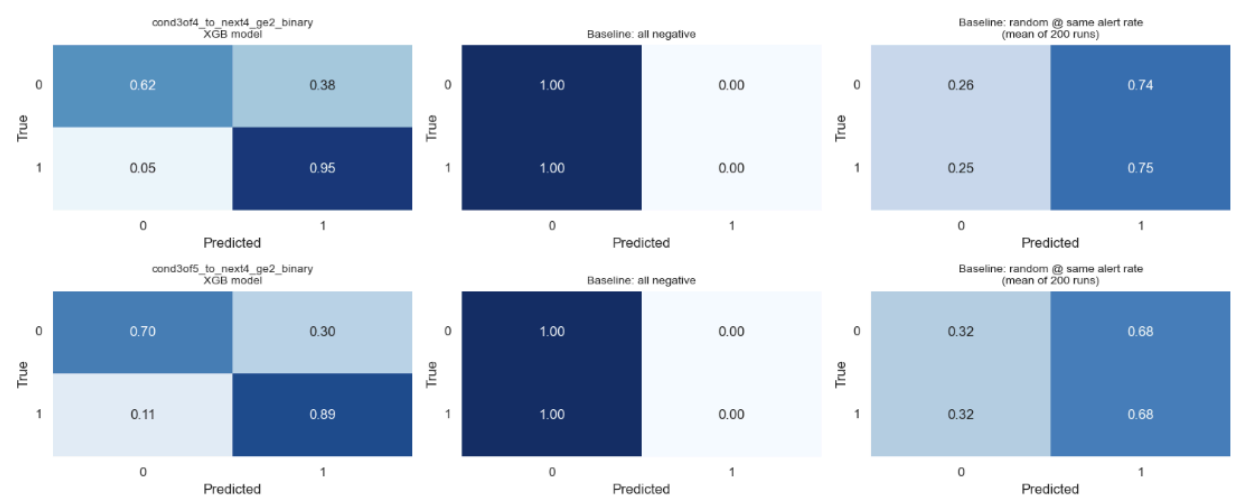
Finally, all current bunching episodes with eligibility flags, can be used as the input vector X and risk-event probabilities can be calculated from eligible episodes, together with features and lagged features associated to that episode.

Modeling overview

For each risk-event type, we use **XGBoost** as our model for our binary classification prediction task. For training and testing purposes, we use a time-sensitive train/validation/test split our of data set. The splits are given by:

- Training set: 2024/07/01 - 2025/09/30
    - A single buffer day
- Validation set: 2025/10/01 - 2025/11/15
    - A single buffer day
- Test set: 2025/11/16 - 2026/03/11

We do a 4-fold cross-validation scheme for model training. A hyperparameter grid search is done to search for optimal XGboost hyperparameters. Depending on the skewed-ness of the specific type of risk-event, we may choose different metrics for assessing performance on validation sets. Generally speaking, as we are leaning towards recall rate, we consider F_2 -score, with possibly a cap on false positive rate.

Model performance results

A sample of inputted risk-event types were tested in order to examine performance. For example, a risk-event of type (3,4,2,4) to be interpreted as: **"given 3 bunches in last 4 stops, at least 2 bunches will occur in the next 4 stops"** resulted in a recall rate of 95.21%, a precision rate of 82.2% and a false positive rate of 36%. For the risk-event of type (3,3,1,2), the corresponding model performed with a recall of 96.7%, precision of 87.6% and false positive rate of 41%. Below are confusion matrices for specific risk-event classification tasks compared to two baselines, after some tuning.